



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 06-301-1

Date: 14-12-2021

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**TECHNICAL SPECIFICATION
FOR
SINGLE CORE PVC- INSULATED STRANDED NON SHEATHED
CABLES (WIRES) OF RATED VOLTAGES UP TO AND INCLUDING
450/750 V**



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Issue: DEC-2021 / Rev- 1

- The revision contains option items that must be selected by ...EDC befor bidding



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1. SCOPE

This Specification covers the minimum technical requirement for design, engineering, manufacturing, inspection, testing and performance of PVC insulated single core stranded non sheathed cables (wires) rated up to 450/750V, suitable for direct burial, installation in ducts and for general purpose.

2. DEFINITIONS

a) Polyvinyl chloride compound (PVC):

Combination of materials suitably selected, proportioned and treated, of which the characteristic constituent is the elastomer polyvinyl chloride or one of its copolymers. The same term also designates compounds containing both polyvinyl chloride and certain of its polymers

b) Rated voltage:

The rated voltage of a cable is the reference voltage for which the cable is designed and which serves to define the electrical tests

The rated voltage is expressed by the combination of two values U_0/U , expressed in volts:

U_0 being the r.m.s. value between any insulated conductor and "earth" (metal covering of the cable or the surrounding medium);

U being the r.m.s. value between any two-phase conductors of a multicore cable or of a system of single-core cables.

In an alternating current system, any rated voltage of a cable shall be at least equal to the nominal voltage of the system for which it is intended

This condition applies both to the value U_0 and the value U

Note: The operating voltage of a system may permanently exceed the nominal voltage of such a system by 10%. A cable can be used at a 10% higher operating voltage than its rated voltage if the latter is at least equal to the nominal voltage of the system



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c) Test voltage:

Unless otherwise specified, the test voltages shall be a.c. 49 Hz to 61 Hz of approximately sine-wave form, the ratio peak value/r.m.s. value being equal to $\sqrt{2}$ with a tolerance of $\pm 7\%$. The values quoted are r.m.s. values.

d) Type tests (symbol T):

Tests required to be made before supplying a type of cable covered by the standard on a general commercial basis in order to demonstrate satisfactory performance characteristics to meet the intended application. These tests are of such a nature that, after they have been made, they need not be repeated unless changes are made in the cable materials or design which might change the performance characteristics.

e) Sample tests (symbol S):

Tests made on samples of completed cable or components taken from a completed cable, adequate to verify that the finished product meets the design specifications.

f) classification:

The conductors have been divided into 4 classes, 1,2,5,6.

Those in class 1 and 2 are intended for use in cables for fixed installations.

Classes 5 and 6 are intended for use in flexible cables and cords but also may be used for fixed installations.

Class 1: solid conductors.

Class 2: stranded conductors.

Class 5: flexible conductors.

Class 6: flexible conductors which are more flexible than class 5.

3. APPLICABLE CODES AND STANDARDS

The latest revision of the following Codes and Standards shall be applicable for the cables covered in this specification. In case of conflict, the vendor/manufacture may propose



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equipment/material conforming to one group of Codes and Standards quoted hereunder without jeopardizing the requirements of this Specs:

Table (1)

S.N	Standard No.	Description
1	IEC 60228	Conductors of Insulated Cables.
2	IEC 60811	Test Methods for non-metallic materials.
3	IEC 60227series	Polyvinyl Chloride Insulated Cables Of Rated Voltages Up to And Including 450/750V: Part 1: General Requirements. Part 2: Test Method. Part 3:Non-Sheathed Cables for fixed wiring.
4	ES 2948	Egyptian Standard for Conductors of Insulated Cables.
5	ES 182 Series	Egyptian Standard for Polyvinyl Chloride Insulated Cables Of Rated Voltages Up to And Including 450/750V

4. ENVIRONMENTAL CONDITIONS

The performance of the stranded cables guaranteed for following environmental conditions, any differences in the guaranteed performance should be clearly set out in the offer.

Table (2)

Minimum ambient temperature	-5°C
Maximum ambient temperature	45°C (50 °c as option).
Maximum relative humidity	95%.
Maximum altitude	1000 m.
Maximum conductor temperature in normal use	70°C.



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5. DESIGN AND CONSTRUCTION REQUIREMENTS

5.1 General

5.1.1 Cables shall meet or exceed the requirements of this Specification in all respects. All cables shall be free of material and manufacturing defects, which would prevent it from meeting the requirements of this Specification.

5.1.2 Manufacturer's drawings shall show the outline of the cables, together with all pertinent dimensions. Any variations in these dimensions due to manufacturing tolerances shall be indicated.

5.2 Design Criteria

5.2.1 Unless otherwise specified, the cables shall be manufactured and tested in accordance with the referred standards.

5.3 Materials

5.3.1 Conductor

- The conductor shall consist of one of the following:
 - Copper or aluminum according to ...EDC requirements.
 - Stranded circular or compact according to ...EDC requirements.
- The conductor shall be class 2 as per IEC 60228 and ES 2948.
- Stranded circular or compact aluminum conductors shall have a cross section area not less than 10 mm².



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Table 3-Stranded conductor for single core.

Nominal c.s.a mm ²	Min NO of wires in the conductor				Max. DC resistance of conductor at 20°C Ω/km	
	Circular		Circular compacted		Annealed CU. Conductor	AL conductor
	CU	AL	CU	AL	Plain wires	
1.5	7	-	6	-	12.1	-
2	7	-	6	-	9.14	-
2.5	7	-	6	-	7.41	-
3	7	-	6	-	6.13	-
4	7	-	6	-	4.61	-
6	7	-	6	-	3.08	-
10	7	7	6	6	1.83	3.08
16	7	7	6	6	1.15	1.91
25	7	7	6	6	0.727	1.20
35	7	7	6	6	0.524	0.868
50	19	19	6	6	0.387	0.641
70	19	19	12	12	0.268	0.443
95	19	19	15	15	0.193	0.320
120	37	37	18	15	0.153	0.253
150	37	37	18	15	0.124	0.206
185	37	37	30	30	0.0991	0.164
240	37	37	34	30	0.0754	0.125
300	61	61	34	30	0.0601	0.100
400	61	61	53	53	0.0470	0.0778

5.3.2 Insulation

Stranded cables insulation shall be polyvinylchloride (PVC). Insulation shall comply with the appropriate requirements specified in IEC 60227-3 and ES 182-3. The nominal insulation thickness shall be as per values specified in IEC60227-3 and ES 182-3, with respect to voltage rating and cables cross-section. The nominal minimum insulation thickness of stranded cables shall be according to relevant standard. The average insulation



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thickness shall not be less than the specified nominal value. The minimum thickness of the insulation at any point shall not fall below the nominal value by more than $0.1 \text{ mm} + 10\%$ of the specified nominal value

Type of insulation will be PVC/C for fixed installation

- 1- Rated voltage: 450/750V
- 2- Code designation 60227 IEC01.

Table 4-Stranded insulation for single core cable.

Nominal c.s.a mm ²	Thickness of insulation mm	Mean over all diam. Mm		Min. AC insulation resistance at 70°C MΩ/km
		Lower limit	Upper limit	
1.5	0.7	2.7	3.3	0.010
2	0.8	3.1	3.8	0.009
2.5	0.8	3.3	4	0.009
3	0.8	3.5	4.3	0.0085
4	0.8	3.8	4.6	0.0077
6	0.8	4.3	5.2	0.0065
10	1	5.6	6.7	0.0065
16	1	6.4	7.8	0.0050
25	1.2	8.1	9.7	0.0050
35	1.2	9.0	10.9	0.0043
50	1.4	10.6	12.8	0.0043
70	1.4	12.1	14.6	0.0035
95	1.6	14.1	17.1	0.0035
120	1.6	15.6	18.8	0.0032
150	1.8	17.3	20.9	0.0032
185	2	19.3	23.3	0.0032
240	2.2	22.0	26.6	0.0032
300	2.4	24.5	29.6	0.0030
400	2.6	27.5	33.2	0.0028



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6. Marking

All cables shall be marked according to IEC60227-1:

- a. The manufacturer's name
 - b. Voltage designation
 - c. Type of insulation, PVC
 - d. Conductor size and material
 - e. Year of manufacture
 - f. Cumulative length at every one meter with the highest length mark on the outer end of the cable only for reel.
 - g. Customer NameEDC
- The marking must be repeated at least every 3 m and the length of wire shall be marked every one meter length.
 - Color of insulation May be (red – blue – Yellow – Black – Green Yellow as required) as option according toEDC requirements.

7. TESTS

7.1. General

7.1.1 All cables shall be tested in accordance with the latest standards and as specified herein. Supplier shall provide all test results for review and acceptance by ...EDC.

7.1.2 The full range of Sample and Type tests specified in latest version of IEC 60227-3.

7.1.3 Sample tests shall be carried out in the supplier's factory according to table 2 of IEC 60227-3. Type test report/certificate shall be submitted toEDC.

7.2. Sample test(S)

Following Sample tests shall be carried out on all finished cables:



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- 7.2.1. Electrical Resistance of Conductors, resistance values shall be in accordance with relevant IEC.
- 7.2.2. AC voltage test, cable shall be tested at test voltage 2500 V. Values of the test voltages for 5 minutes.
- 7.2.3. Checking of compliance with constructional previous
- 7.2.4. Measurement of insulation thickness
- 7.2.5. Measurement of overall diameter.



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7.3 Type Tests

Table 5-Complete test shall be carried out as per the relevant IEC 60227-3.

1	2	3	4	
Ref. No.	Test	Category of test	Test method described in IEC	Subclause
1	<i>Electric test</i>			
1.1	Resistance of conductors	T, S	60227-2	2.1
1.2	Voltage test at 2 500 V	T, S	60227-2	2.2
1.3	Insulation resistance at 70 °C	T	60227-2	2.4
2	<i>Provisions covering constructional and dimensional characteristics</i>		60227-1 and 60227-2	
2.1	Checking of compliance with constructional provisions	T, S	60227-1	Inspection and manual tests
2.2	Measurement of insulation thickness	T, S	60227-2	1.9
2.3	Measurement of overall diameter	T, S	60227-2	1.11
3	<i>Mechanical properties of insulation</i>			
3.1	Tensile test before ageing	T	60811-1-1	9.1
3.2	Tensile test after ageing	T	60811-1-2	8.1.3.1
3.3	Loss of mass test	T	60811-3-2	8.1
4	<i>Pressure test at high temperature</i>	T	60811-3-1	8.1
5	<i>Elasticity and impact strength at low temperature</i>			
5.1	Bending test for insulation	T	60811-1-4	8.1
5.2	Elongation test for insulation ¹⁾	T	60811-1-4	8.3
5.3	Impact test for insulation	T	60811-1-4	8.5
6	<i>Heat shock test</i>	T	60811-3-1	9.1
7	<i>Test of flame retardance</i>	T	60332-1	

¹⁾ Only applicable if the overall diameter of the cable exceeds the limits specified in the test method.

8. PACKING AND SHIPPING

The packing and shipping of the cable shall conform to the following:

8.1. The cable ends shall be sealed with a waterproof, elastomeric end cap for drum

8.2. The cable shall be sealed with heat shrinkable plastic for coils.



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8.3. Unless ...EDC mention otherwise, Cables shall be supplied in lengths of 100 meters per roll (up to 25 mm²) and it should be well sealed against dust, water and weather conditions.

8.4. Unless ...EDC mention otherwise, Cables shall be supplied on reels for C.S.A ≥ 35 mm². With the following data shall be printed on both sides:

ForEDC:

Manufactured by:

Rated voltage: V

C.S.A of the cables: mm²

Material of conductor: CU/AL

Type of insulation: PVC

Length of cable: M

Cross weight: KGs

Net weight: KGs

Reel No./Date of manufacture:

9. Guarantee

- The supplier should guarantee the Cable against all defects arising out of faulty designed or workmanship or of defective materials for a period of 12 months from putting in service, or 18 months after delivery, whichever expires earlier.
- The good quality and manufacturing of the materials should be guaranteed.
- Any parts which on account of poor quality of the materials, manufacturing defects or poor assembly, should be replaced or repaired in the shortest time possible and free of charge.



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10. Drawing and Catalogues:

- All necessary drawings of the cable construction, Technical catalogues containing the technical data and reference standards applied to the cables and instructions for erection, operation, quality control procedure and maintenance should be submitted with the offer.
- Any tender not accompanied with clear and complete guarantee tables should be rejected.
- Type test report/certificate shall be submitted toEDC.

11. Guarantee Table

- The tenderer should fill in the attached guarantee tables.
- Any tender not accompanied with clear and complete guarantee tables should be rejected



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GURANTEE TABLE
FOR CU. /AL CABLESmm²

-	Manufacturer name	
-	Standards to which cables are designed and tested	
-	Rated voltage	v
-	Nominal cross-sectional area of conductors	mm ²
-	No of wires / Diameter of each	mm
-	Max. resistance of conductor at 20°C	Ω/km
-	Insulation thickness	mm
-	Mean overall diameter	mm
-	Min insulation resistance at 70°C	M Ω .km
-	Weight of CU/AL	Kg/m
-	Gross weight	Kg/m

I/We certify the correctness of the information given above for of cu. cables offered for --
....EDC.

Date: / /

Signature of Tenderer